



AI And Machine Learning

The Case for Using Small Language Models

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Summary. Small Language Models (SLMs) are redefining enterprise AI by offering faster, more efficient, and cost-effective solutions compared to Large Language Models (LLMs). Their compact design enables deployment on edge devices, allowing real-time decision-making... [more](#)

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Until now, the AI revolution has been largely measured by size: the bigger the model, the bolder the claims. However, as we move closer to truly autonomous and pervasive AI systems, a new trend is emerging that suggests a surprising shift: smaller may actually

be better. According to a [recent study by NVIDIA researchers](#), [small language models \(SLMs\)](#), rather than their larger counterparts [large language models \(LLMs\)](#), could become the true backbone of the next generation of intelligent enterprises. The age of “bigger is better” may be giving way to “smaller is smarter.”

SLMs have many competitive advantages over LLMs. Their architectures ([Microsoft’s Phi-3](#), [Google’s Gemma](#), [Apple OpenELM](#), [IBM Granite](#)) are more compact, specialized, energy-efficient, and easier to deploy than LLMs, especially on local edge computing devices. SLMs have significantly fewer parameters, typically ranging from a few billions, compared to LLMs, which can contain hundreds of billions or even trillions. This substantial difference in size provides several [advantages](#) for SLMs over LLMs, including lower computational requirements, faster training times, easier deployment, and more efficient performance in specific scenarios.

In this article we explore why SLMs are often better suited for many AI use cases, including agentic AI applications, and what business leaders should consider as they shape their AI strategies.

Speed and Efficiency Drive Competitive Advantage

SLMs are lightweight enough and can be deployed locally on edge devices or local servers (e.g., smartphones, wearables, factory sensors, or even home appliances), and can offer faster response times. Unlike LLMs, which require cloud-based inference and massive computation, SLMs can deliver real-time decision-making at the point of need. For applications such as voice

assistants (e.g., Siri, Alexa, Google Assistant), autonomous vehicles, smart Industry 5.0 factories, online gaming agents, AI-assisted remote surgery, and wearable health monitors, where the response time measured in milliseconds rather than seconds can make or break the user experience, SLMs can offer a strategic advantage by reducing latency and enhancing reliability. Moreover, the compact size of SLMs makes them ideal for edge computing, enabling agentic AI to operate without relying on cloud connectivity. For example, several companies now offer smartphone-based SLMs that can do real-time language translation. Because they don't rely on the cloud, response time is much faster.

Running LLMs at scale is both costly and energy-intensive-not only during training phase but also during ongoing operation. Using them continuously in autonomous agents is often financially unsustainable. In contrast, SLMs are significantly more energy-efficient and cost-effective. They can be trained and deployed using just 30–40% of the computational power required by their larger counterparts LLMs, making widespread adoption across edge devices and smaller enterprises far more feasible. This shift is important not just for reducing operational costs, but also for supporting companies' sustainability goals. In sectors where AI needs to be deployed across thousands of devices or end users, SLMs can fundamentally change the economics of deployment.

Specialization can Outperform Generalization

LLMs are like generalists because they are trained on massive amounts of internet data, and they can address almost any topic. SLMs, on the other hand, can be fine-tuned for specific tasks or industries, like healthcare, legal work, or managing supply chains. Because they are trained on a narrower range of data,

SLMs are often more reliable for domain-specific agentic tasks and are less likely prone to hallucinate or give off-topic answers.

Companies need not necessarily fine-tune their own SLMs. Some vendors are beginning to develop SLMs that are targeted to particular industries or business problems. Infosys, for example, has developed SLMs that are optimized for banking and for IT operations. They can be used as the basis for further customization on the client's content.

Because of their focused training data, SLMs are also faster and more cost-effective to build, fine-tune, and improve over time than LLMs. These qualities of SLMs can give businesses the agility to develop AI solutions that can quickly adapt to changing market dynamics, shifting customer expectations, or new regulatory demands without the lengthy development cycles and high sunk costs typically associated with LLMs.

Bayer, for example, has developed an SLM to help its front-line personnel answer farmers' questions about crop protection called E.L.Y, for "Expert Learning for You." It is trained on Bayer's crop protection knowledge and was found to be 40% more accurate than initial testing with a large model. It can also be customized to more specific crops or geographical farming knowledge.

Agentic AI with More Control and Better Privacy

Because agentic AI involves greater autonomy, SLMs can provide a higher level of control and trust. In agentic applications where predictability, transparency, and auditability are more important than raw computational power, an agent trained on a SLM, with fewer parameters and well-defined training data, is easier to monitor, test, audit, and control.

SLMs can operate entirely on an end-user's mobile device or laptop, enabling companies to process sensitive information such as financial records, health data, employee records, legal documents, and intellectual property without sending it to the cloud for storing and further processing.

This capability is especially very important in industries like defense, finance, healthcare, and transportation, where safety, ethics, and regulatory compliance are major concerns. In these sectors, explainability is non-negotiable, transparency is a strategic asset, and SLMs can play a key role in building customer trust and ensuring adherence to data protection regulations like GDPR. For agile organizations, SLMs can effectively support the “build-measure-learn (BML)” cycle in agentic AI innovation, enabling rapid experimentation and continuous improvement.

Safer for Prototyping, Experimentation, and Embedding in Workflows

Because SLMs are lower-risk and easier to control, they can be well-suited for low-stakes experimentation. This can reduce barriers to innovation and promote a culture of safe, hands-on experimentation, where teams can test their ideas without high risk or substantial investment. Organizations can quickly prototype new agentic workflows like automating procurement tasks or streamlining HR onboarding, managing IT helpdesk requests, handling expense approvals, scheduling maintenance operations, or tracking inventory levels, all without needing large-scale infrastructure or relying on external vendors.

Unlike LLMs, which often demand substantial infrastructure and major system upgrades, SLMs can be seamlessly integrated into existing enterprise software, APIs, or edge devices without the need for major architectural changes. This makes SLMs ideal for gradual, “inside-out” transformation, where businesses enhance

legacy systems rather than replace them. This becomes easier to understand if we imagine AI assistants embedded within ERP platforms, CRM tools, or mobile apps, instead of building completely separate AI layers for a specific task.

What Executives Should Do with SLMs

For business leaders, this growing shift towards SLMs can carry significant strategic implications. Instead of assuming that “bigger is better” in AI, it’s time to rethink strategy. As organizations are moving beyond experimentation and into deployment at scale, it is clear that the smartest systems are not necessarily the biggest and that the future of intelligent enterprise systems will be driven not by powerful AI models, but by the right-sized AI models that are designed with specific business contexts.

Many situations will clearly point to smaller models than large. In a moving vehicle, for example, connectivity and real-time response needs will often favor SLMs over large models. For example, Cerence AI, a provider of mobile systems, has worked with Microsoft to develop an SLM for various automotive uses, including interacting with infotainment and heating/air conditioning controls. If their AI roadmap had been focused solely on LLMs, the technology may not have fit the mobile context and constraints. A narrow focus on LLMs alone can prevent an organization from gaining a meaningful competitive edge, limiting its ability to achieve faster deployments, reduce costs, enhance system safety, and ultimately deliver smarter, more effective enterprise applications.

To stay ahead, forward-thinking AI leaders must begin asking the right strategic and operational questions within their organizations that are developing generative and agentic AI systems:

- Are we matching the size of the generative AI model to the complexity and constraints of the task?
- Are we identifying the right AI use cases within the organization, particularly those requiring real-time, localized, or privacy-sensitive decision-making?
- How can SLMs offer unique advantages or mitigate risks associated with relying solely on LLMs?
- Are our generative and agentic AI investments aimed at long-term scalability, or simply short-term experimentation?
- How will we govern, monitor, and control SLM-powered agents, especially those running outside the cloud on local edge devices?
- How can we invest to build domain-specific SLMs that are fine-tuned for our unique business processes, rather than relying solely on general-purpose LLMs?
- How can we rethink our agentic AI infrastructure to enable SLMs to run at the edge, offering faster response times, stronger privacy protections, and scalable deployment while ensuring alignment with our organization's values and long-term strategic goals?
- How will we monitor and evaluate the performance and KPIs of our SLMs in production, and how will we manage their lifecycle covering updates, retraining, and version control?
- What ethical considerations must we address with our SLM deployments, which regulatory compliance frameworks (e.g., GDPR, HIPAA) apply, and how will we ensure adherence to them?

While LLMs have shown remarkable capabilities in recent years, organizations should open the doors to distributed intelligence by prioritizing the development of SLMs. SLMs are more cost-effective, easier to deploy on local infrastructure, and better suited for fine-tuning with proprietary or domain-specific data. In many cases they bring greater efficiency, accessibility, and trust. They offer faster response times, lower energy consumption, and better alignment with data privacy and regulatory compliance requirements.

As the adoption of agentic AI accelerates, the agility and adaptability of SLMs will establish them as a strategic asset for organizations seeking scalable, responsible, and context-aware AI solutions. Overall, SLMs may not astound users like their larger counterparts, but they will quietly drive bottom-line impact in the years ahead. Business leaders should pay close attention to these models, identifying business processes where lightweight, task-specific SLMs can deliver quick wins, and preparing to invest strategically.



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